

Question 1: Foundations of Machine Learning [overall 34 marks]

There are many different types of credit card fraud, but at its simplest, it's when someone obtains your card details and makes transactions on your card without you knowing.

- a) Explain what the credit card fraud transaction problem is in the context of machine learning for this simplest scenario and what T, P & E are?

[7 marks]

(1 mark) definition

(6 marks) P, E & T

Sample answers:

Definition: A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

A computer program is said to learn from credit card transaction experience E with respect to fraud detection task T and detection accuracy performance measure P, if its performance on detection task T, as measured by accuracy performance P, improves with credit card transaction experience E.

P, E & T:

Task T: credit card fraud detection

Performance Measure P: percentage of instances of credit card transaction correctly predicted

Training Experience E: a database of credit card transaction records with different factors/attributes which will affect the fraud detection

- b) Give the formal definition of overfitting. Explain when the overfitting occurs and why overfitting is such a big problem for Machine Learning?

[5 marks]

(2 marks) Definition

(3 marks) Problems+Why

Sample answers:

Given a hypothesis space $\mathcal{H}$, $h \in \mathcal{H}$ overfits the training data if $\exists h' \in \mathcal{H}$ such that h has a smaller error over the training examples, but h' has a smaller error over the entire distribution of instances.

Overfitting can occur when:

Learning is performed for too long (e.g. in Neural Networks)

The examples in the training set are not representative of all possible situations (called predictor bias, and is usually the case).

Model parameters are adjusted to uninformative features in the training set that have no causal relation to the true underlying target function!

This is a major problem for machine learning as it basically results in a high generalisation error, and the whole purpose of ML is to reach a low generalisation error.

- c) We trained two classifiers, Logistic Regression with 300 examples and tested with 30 examples; and the Support Vector Machine with 400 examples and tested with 50 examples. The correctly classified percentage of the Logistic Regression is 82% and for the support vector machine it is 92%. Calculate the confidence intervals of the

two true errors and comment how that relates to the statistical difference between the results.

$Z_n = 1.96$ for a confidence level of 95%.

[6 marks]

(2 marks) $\hat{d}, n$ value

(2 marks) interval

(2 marks) comments

Sample answers:

$$\hat{d} \equiv e_{S_1}(h_1) - e_{S_2}(h_2)$$

$$\hat{d} \equiv 0.18 - 0.08 = 0.1$$

$$n_1 = 30; n_2 = 50$$

$$\hat{d} \pm Z_n \sqrt{\frac{e_{S_1}(h_1) (1 - e_{S_1}(h_1))}{n_1} + \frac{e_{S_2}(h_2) (1 - e_{S_2}(h_2))}{n_2}}$$

$$[0.1 \pm 1.96 \sqrt{\frac{0.18 (1 - 0.18)}{30} + \frac{0.08 (1 - 0.08)}{50}}]$$

we can now say that with 95% probability that the true error difference between Logistic Regression and support vector machine will lie in the interval $[0.26, -0.06]$.